

Solari

Solar Industry Chatbot

User Guide & Evidence of Implementation

Project Documentation - Track 2 Submission

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Section owned: User Guide · Screenshots & Evidence of Implementation
Solar Industry Chatbot - Internship Group Project

1. Introduction

Solari is a web-based solar industry chatbot designed to help users understand solar panel systems, get product comparisons, receive an approximate system-sizing recommendation, and access supporting research documents - all through a simple conversational interface. This guide explains how to access the application and use every feature of the interface from an end user's point of view.

This document covers:

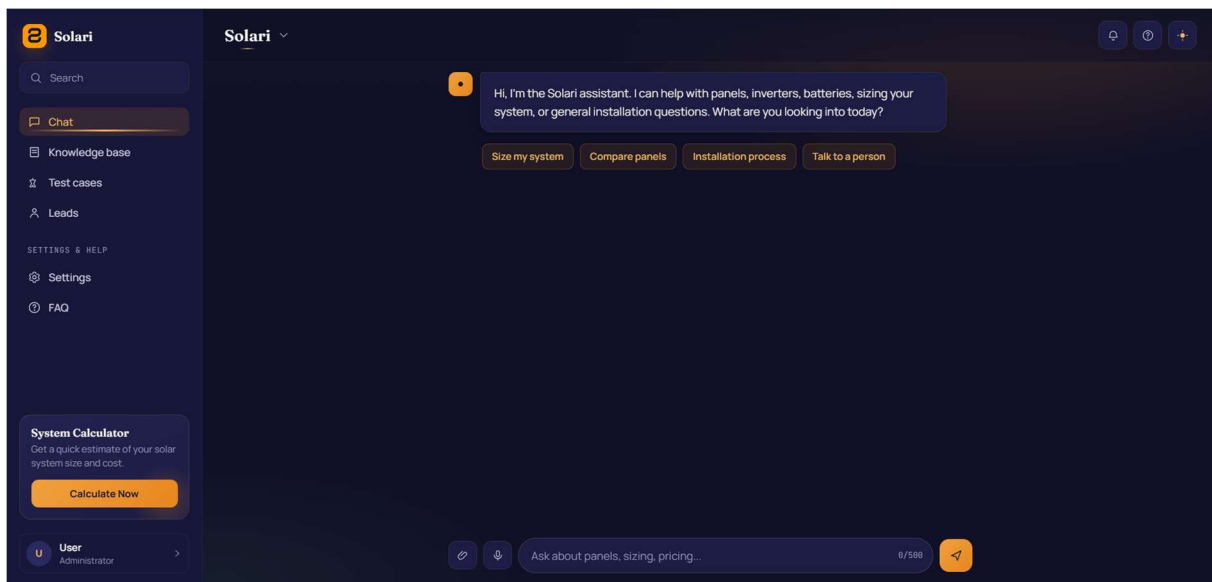
- Accessing and launching Solari
- A walkthrough of the interface layout
- How to start and hold a conversation with the assistant
- Using the built-in Solar System Calculator
- Browsing the Knowledge Base and Test Cases sections
- Switching themes and understanding on-screen indicators
- Troubleshooting common issues
- Screenshots and evidence of the working implementation

2. Accessing the Application

Solari runs as a browser-based interface backed by a Python API server. To use the application:

1. Ensure the backend server is running (see the Installation & Setup section prepared by Zain).
2. Open frontend/index.html in a web browser, or navigate to the URL where the frontend is hosted.
3. The chat interface loads automatically with a welcome message from the assistant.

No login or account creation is required - the interface opens directly into an active chat session.



S1 - Landing / Welcome Screen

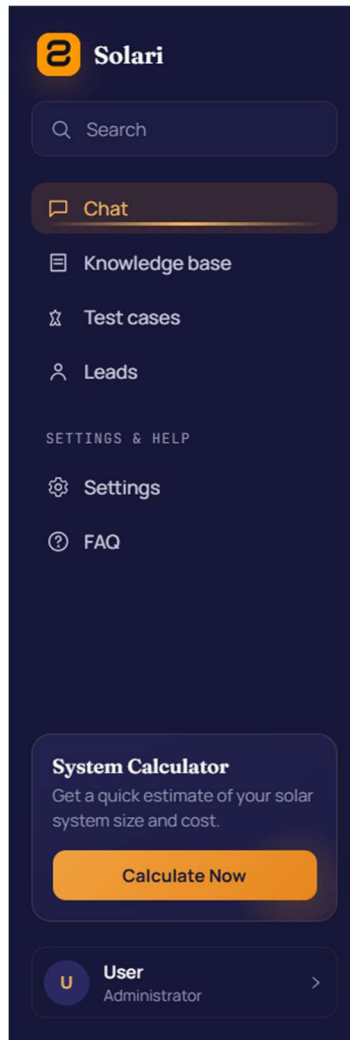
Full browser window showing Solari immediately after opening index.html, with the welcome message and suggestion chips visible.

3. Interface Overview

The Solari interface is divided into a left navigation panel and a main content area.

3.1 Left Navigation Panel

- Brand header - displays the Solari logo and name.
- Search box - a field to search within the app (UI element).
- Primary navigation - three tabs: Chat, Knowledge base, and Test cases.
- Secondary navigation - Settings and FAQ links.
- System Calculator promo card - a shortcut button to open the sizing calculator.
- Profile card - displays the current user and role at the bottom of the panel.



S2 - Left Navigation Panel

Close-up of the sidebar showing the logo, search box, nav links (Chat / Knowledge base / Test cases), and the System Calculator promo card.

3.2 Main Chat Area

- Header - shows the assistant name and quick-access icons for notifications, help, and theme toggle.
- Conversation thread - displays the running exchange between the user and the assistant.
- Suggestion chips - quick-start prompts (e.g., "Size my system", "Compare panels", "Installation process", "Talk to a person").
- Message composer - the text input where the user types a question, with a live character counter (0/500) and a send button.

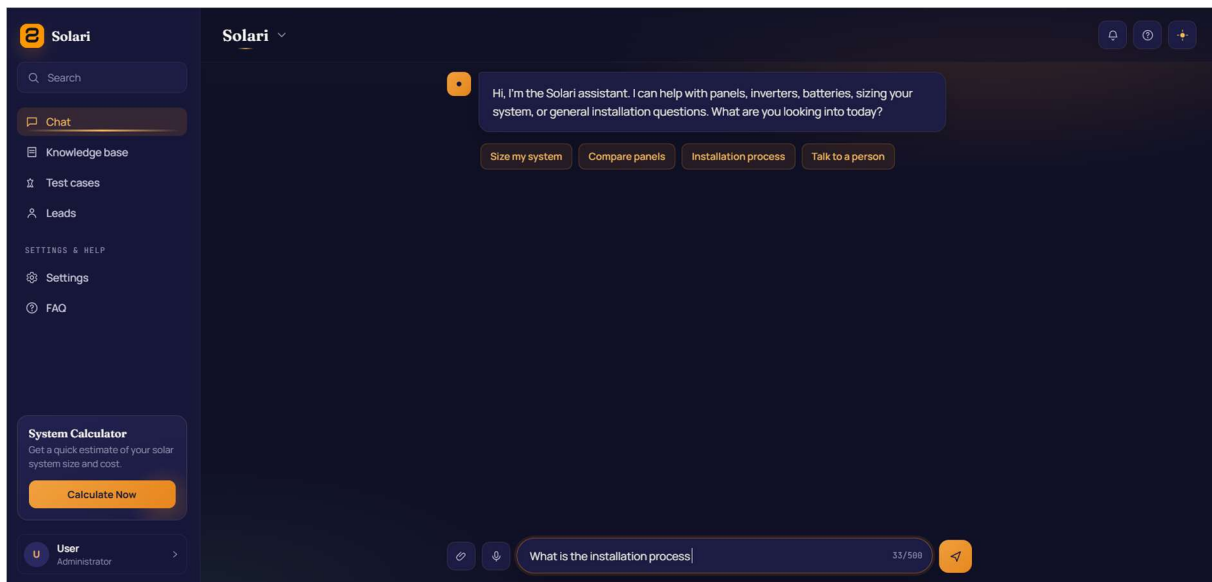
4. Starting a Conversation

There are two ways to begin chatting with Solari:

1. Type a question - click the message box at the bottom of the screen, type a question (up to 500 characters), and press Enter or click the send button.
2. Use a suggestion chip - click one of the pre-written prompts above the composer to instantly send that question.

Once a message is sent:

- The user's message appears as a bubble on the right side of the thread.
- A typing indicator (three animated dots) appears briefly while the assistant prepares a reply.
- The assistant's response appears as a bubble on the left side, formatted with headings, bold text, and lists where relevant.



S3a - Composer / Character Counter in Use

A question typed into the message box before sending, showing the live character counter (33/500).

5. Using the Solar System Calculator

The System Calculator gives users a quick, approximate estimate of the solar system they would need, based on their electricity usage.

1. Click "Calculate Now" on the promo card in the sidebar. This opens the calculator as a pop-up window (modal).
2. Enter the Average Monthly Usage in kWh (required).
3. Optionally enter the Required Backup in hours if battery backup is needed.
4. Click "Calculate Size". The tool briefly shows a loading animation, then returns the estimate.

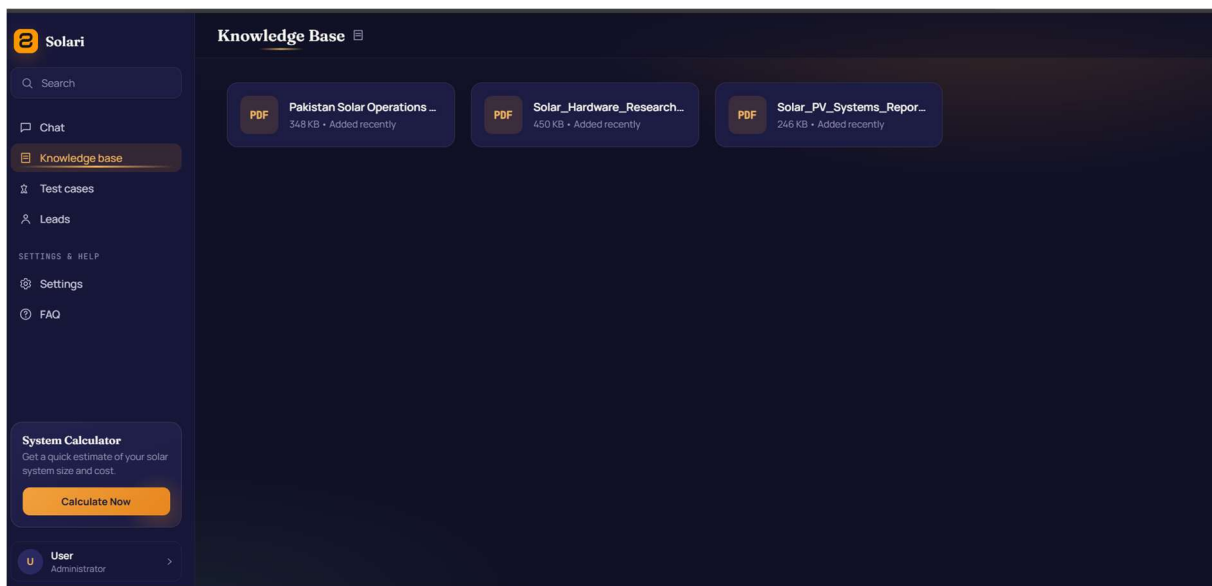
The result includes:

- Recommended system size (kW)
- Number of panels needed
- Estimated roof area required (sq ft)
- Recommended battery backup capacity (kWh), if applicable

The modal can be closed at any time using the close (X) button or by clicking outside the pop-up window.

6. Knowledge Base Tab

The Knowledge base tab lists the reference documents used to ground the assistant's answers. Clicking a document card opens the underlying PDF in a new browser tab. Available documents include research on Pakistan solar operations, solar hardware, and solar PV systems.

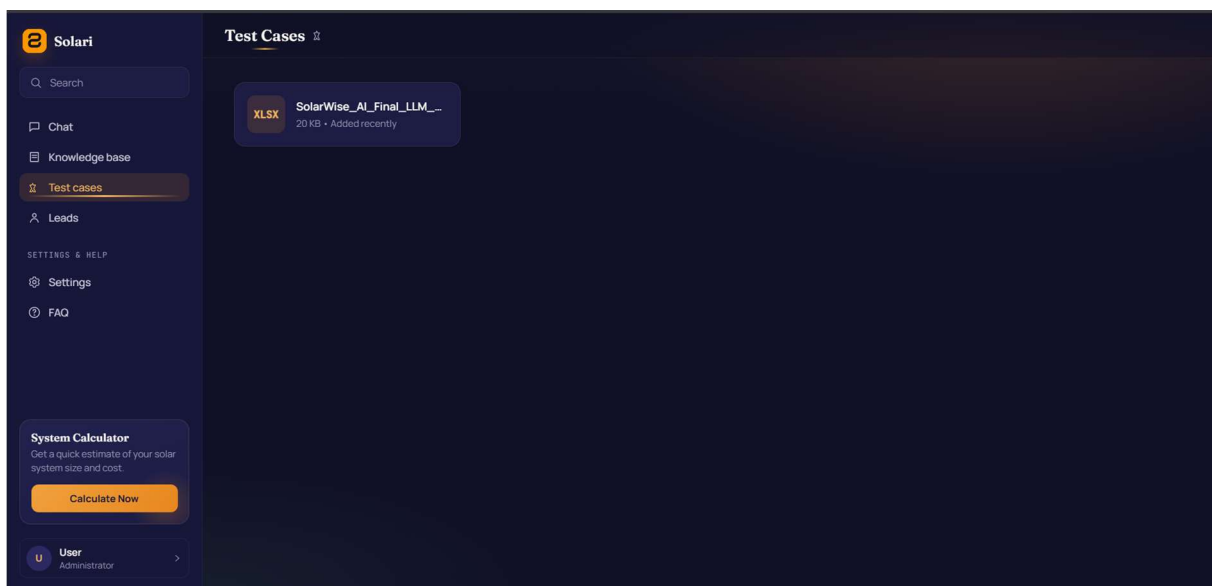


S7 - Knowledge Base Tab

The Knowledge base screen listing the available PDF reference documents as cards.

7. Test Cases Tab

The Test cases tab provides direct access to the project's test documentation - an Excel file containing the prepared test cases used to validate the assistant's responses (compiled separately in the Testing Report by Rauf).



S8 - Test Cases Tab

The Test cases screen showing the linked test case spreadsheet.

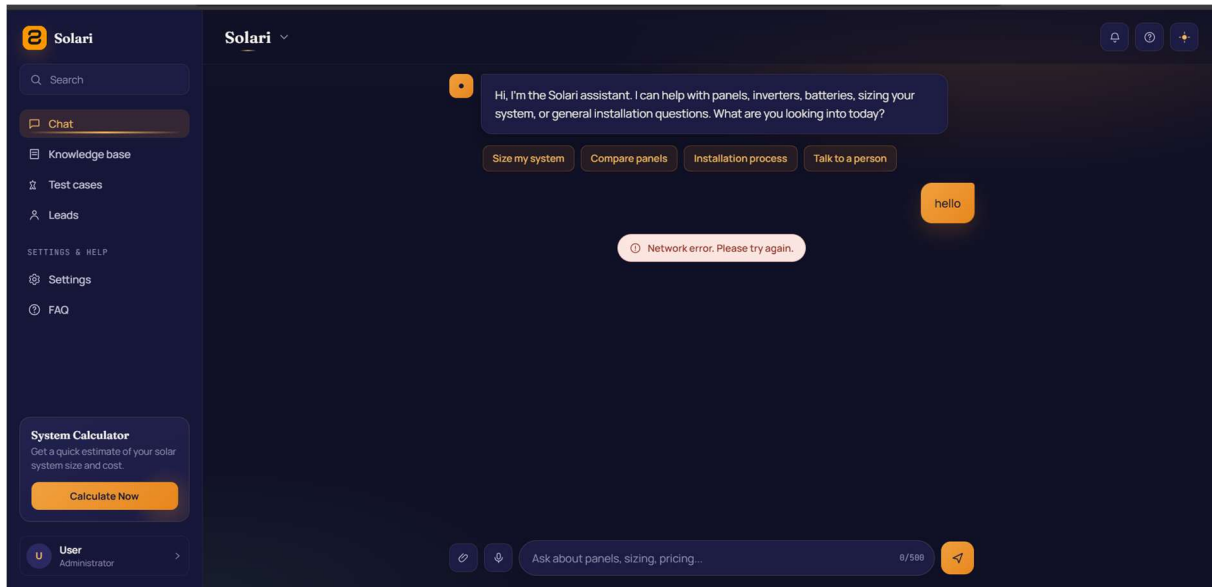
8. Theme Toggle and Other Controls

- Theme toggle (sun/moon icon, top-right of the chat header) - switches the interface between light and dark appearance.

- Character counter - updates live as the user types, capped at 500 characters per message.
- Notifications and Help icons - placeholders in the header for future functionality.

9. Error Handling

If the backend server is unreachable or returns an error while the assistant is generating a reply, a red network error banner appears above the composer: "Network error. Please try again." The banner automatically disappears after a few seconds, and the user can simply resend their message once the connection is restored.



S9 - Network Error Banner

The red 'Network error. Please try again.' banner, captured by temporarily stopping the backend server and sending a message.

10. Troubleshooting

Issue	What to do
Chat does not respond / spins indefinitely	Confirm the backend server (main.py) is running on the expected port, then refresh the page.
"Network error" banner appears	Check that the backend is running and reachable; resend the message once it's back up.
Calculator shows "Failed to calculate recommendation"	Verify both Monthly Usage and Backup Hours are valid numbers, and that the backend /api/recommend endpoint is running.
Knowledge Base / Test Cases links do not open	Confirm the referenced PDF/XLSX files exist in the backend/knowledge_base and backend/test_cases folders.
Message box will not accept more text	The 500-character limit has been reached; shorten the message.

11. Notes on Screenshots & Evidence of Implementation

The screenshots above (S1, S2, S3a, S7, S8, and S9) capture the core screens and states of Solari: the landing chat screen, the navigation panel, an in-progress message, the Knowledge Base tab, the Test Cases tab, and the network error state. Together they provide evidence that the interface described in this guide is implemented and functioning as documented.